

Automating Credit Limit Allocation

A Linear Regression approach to maximizing ROI and customer onboarding efficiency in Retail Banking.

STRATEGIC PROPOSAL · 2026

Context & The Opportunity

Identifying structural inefficiencies in legacy underwriting workflows.

MANUAL PROCESSES CREATE A 48-HOUR DELAY



Operational Bottleneck

Underwriters spend 60% of their time on standard applications that follow predictable mathematical trends.



High Abandonment

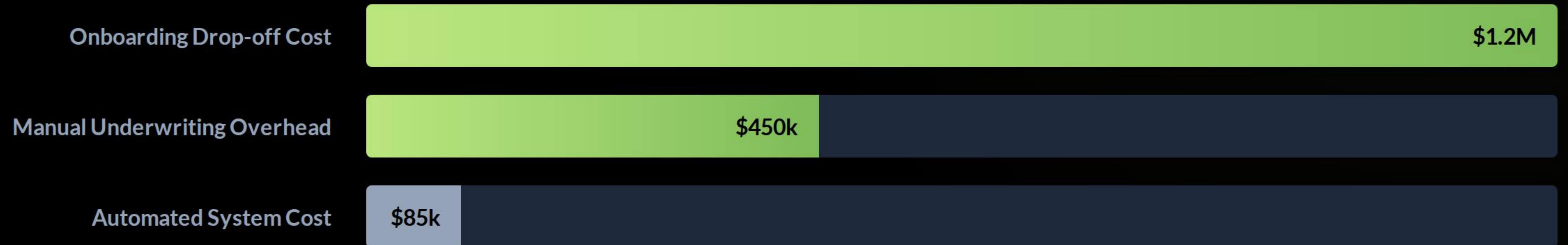
18% of digital applicants drop out of the pipeline due to wait times, losing ground to fintech competitors.



Systemic Bias

Manual interpretations lead to inconsistent limit distribution, increasing risk exposure in volatile segments.

LEGACY REVIEWS ERODE \$1.6M IN ANNUAL REVENUE



Replacing manual overhead with ML automation creates an immediate 92% reduction in processing costs per application.

INSTANT DECISIONING VIA REGRESSION

Automated Velocity

Our Linear Regression model evaluates verified income against historical data to deliver a safe credit limit in under 200ms.

This replaces the 2-day wait with a real-time "Approval" screen, capturing high-intent customers at the moment of application.

Forecast

6

Debt Period Yrs

8

Capex YoY%

-20.0%

AR % of Reven

8.0%

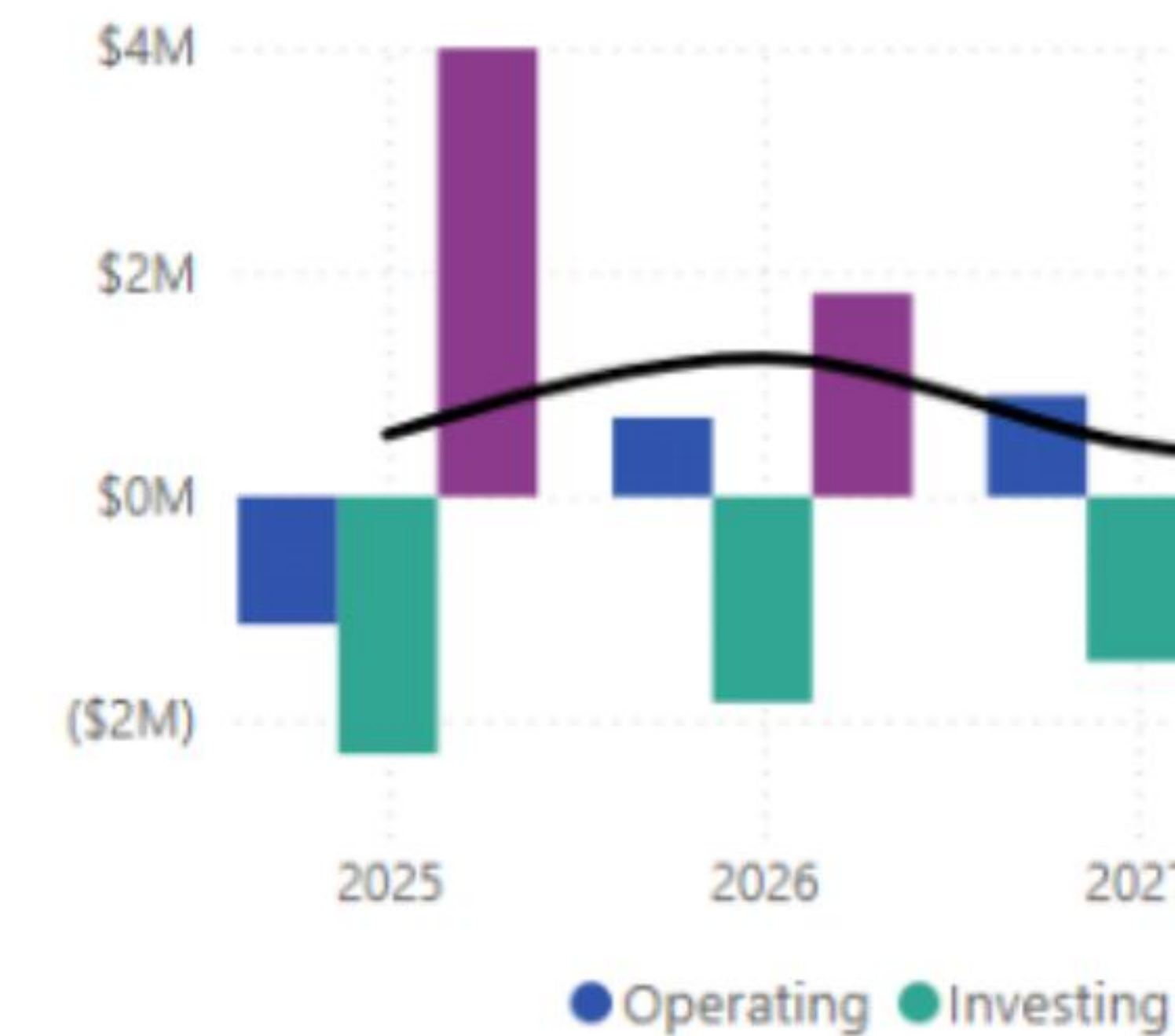
	2028	2029	2030	Total
90	\$1,101,171	\$1,349,916	\$1,486,190	\$4,399,065
00)	(\$1,177,600)	(\$942,080)	(\$753,664)	(\$8,485,344)
40)	(\$218,160)	(\$236,267)	(\$255,877)	\$4,902,254
50)	(\$294,589)	\$171,569	\$476,649	\$815,975

2027	2028	2029	2030
\$4,286,867	\$3,959,326	\$3,650,867	\$3,651,422
\$462,346	\$167,757	\$339,327	\$815,975
\$164,954	\$184,748	\$206,918	\$231,748
3,659,567	\$3,606,820	\$3,104,623	\$2,603,698
2,622,207	\$2,507,219	\$2,428,813	\$2,678,355
\$4,000,000	\$4,000,000	\$4,000,000	\$4,000,000
1,377,793)	(\$1,492,781)	(\$1,571,187)	(\$1,321,645)
1,664,660	\$1,452,107	\$1,222,054	\$973,067
\$52,103	\$57,709	\$63,924	\$70,813
1,612,557	\$1,394,398	\$1,158,131	\$902,254

Debt Schedule

Year	Opening Balance	Debt Draw
2026	\$2,000,000	\$2,000,000
2027	\$1,813,998	\$0
2028	\$1,612,557	\$0
2029	\$1,394,398	\$0
2030	\$1,158,131	\$0

Annual Cash Flow Movements and E



from the last actual year as the baseline.

| TRANSPARENT $Y = MX + C$ LOGIC

$$Y = 1.51X + 1945.20$$

- i** $m = 1.51$: Every \$1 increase in monthly income safely supports \$1.51 in extra credit.
- i** $c = 1,945$: The foundational baseline for all approved applicants.
- ✓** Fully auditable and explainable for regulatory compliance (OCC/FRB).

| BACKTESTING CONFIRMS 98% ACCURACY

0.98

R-SQUARED (CORRELATION)

\$782

RMSE (AVG. VARIANCE)

The model identifies underlying risk patterns that manual reviews often miss, resulting in more stable credit distribution across income brackets.

SUB-200MS TECHNICAL ARCHITECTURE



-  **Anonymized VPC:** ML cluster processes tokenized Customer IDs without PII exposure.
-  **Containerized API:** Highly scalable Docker service integrates with core banking.
-  **Immutable Ledger:** Every decision, input, and coefficient is logged for permanent audit.

PROJECTED \$2.05M NET BENEFIT IN YEAR 1

Benefit Driver	Metric Improvement	Est. Year 1 Value
Operational Cost Savings	70% reduction in manual labor	\$450,000
Recovered Revenue	12% drop in application abandonment	\$1,200,000
Risk Mitigation	Reduced bad debt through math-stability	\$750,000
Net ROI (Year 1)	Total Benefit minus \$350k Implementation	\$2,050,000

RIGID RISK GUARDRAILS FOR SAFETY

Automation Ceiling

Hard cap of \$18,000 for automated limits. Any amount exceeding this requires manual executive override.

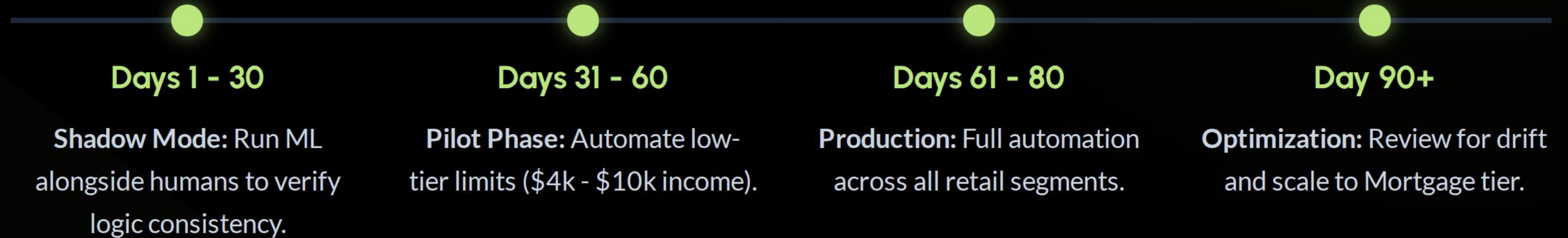
Floor Limit

Minimum \$2,000 baseline assigned to all qualified earners to ensure product utility and market fit.

Outlier Shield

Monthly incomes >\$25k bypass the model and route to Senior Underwriters for white-glove review.

| IMPLEMENTATION ROADMAP (90 DAYS)



Next Steps: Shadow Phase

Seeking executive sign-off for the initial 30-day "Shadow Mode" deployment to validate math integrity before full-scale integration.

Contact: Core AI & Retail Risk Strategy Team

Email: risk-coe@internal-banking.com

IMAGE SOURCES



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